

Zero-Trust Cyber Threat Intelligence (CTI) & Incident Correlation Platform

Continuous Biometric Verification, Optical Edge Sensing, and Autonomic Defenses for Academic Environments

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Backend Stack: Python Flask 3.1 + MySQL 8.0 InnoDB	Telemetry: Keystroke Biometrics, Optical Haar, Audio Vishing
Verification Status: Production Verified (12/12 Automated Tests Passed)	Live Records: 1,002 Students 49 Threat Events

Executive Abstract: Traditional academic campus security architectures rely primarily on static, perimeter-based authentication mechanisms such as initial password prompts or single-sign-on (SSO). Once credentials are exchanged, persistent trust is implicitly granted. This creates critical attack surfaces: in open collegiate computer labs, shared libraries, and dormitories, adversaries easily exploit **Shoulder Surfing**, execute **Session Hijacking** via voice phishing (vishing), or enforce physical duress (**Coerced Session Breaches**).

To solve this systemic vulnerability, this project develops a production-grade **Zero-Trust Cyber Threat Intelligence (CTI) & Incident Correlation Platform**. By uniting a 6-table normalized MySQL database enforcing ACID transactions and referential integrity with an in-memory real-time statistical correlation engine, the system continuously ingests multi-modal client telemetry—namely, millisecond-precision keystroke dynamics (dwell and flight variances), edge optical sensor face counts (OpenCV Haar Cascade), and dynamic social engineering challenge outcomes. The platform implements autonomic incident mitigations (Screen Blur, Session Revocation, and Decoy Honeypot Containment) and maintains an immutable audit trail for security operations.

Core System Capabilities At A Glance:

Component	Implementation Details	Key Security Value
Relational DBMS Core	6 Normalized MySQL InnoDB tables, foreign keys with ON DELETE CASCADE, session indices.	Auditable persistence, ACID compliance, high-speed indexing.
Keystroke Dynamics	Client performance.now() tracking; backend safe population variance engine (σ^2).	Detects emotional duress, typing tempo anomalies, and takeovers.
Optical Edge Sensor	OpenCV Haar Cascade face counting; producer-consumer thread pushing 1 POST/sec.	Identifies unauthorized physical observers in real-time.
Autonomic Defenses	3-tier priority correlation rule engine: Honeypot, Session Revocation, Screen Blur.	Eliminates human response delay by neutralizing threats instantly.

1. Relational Database Architecture & Schema Design

At the foundation of the platform is a normalized relational schema implemented on **MySQL InnoDB** (database name: `vit_cyber_project`). The database enforces 3rd Normal Form (3NF), referential integrity with cascading constraints, dedicated session indexing for sub-millisecond retrieval, and strict data types to ensure zero data corruption.

Table Name	Primary Key & Foreign Keys	Key Columns & Types	Role & Integrity Rules
students	PK: student_id (INT AI) UQ: oauth_email	oauth_email VARCHAR(255) branch VARCHAR(100) year_of_study INT primary_device VARCHAR(50) session_id VARCHAR(100)	Core identity ledger. Uses ON DUPLICATE KEY UPDATE to allow non-destructive student upserts.
survey_responses	PK: response_id (INT AI) FK: student_id → students(student_id)	uses_2fa TINYINT(1) software_update_freq VARCHAR(50) overall_risk_score INT vishing_passed TINYINT(1)	1-to-many relationship tracking cyber hygiene answers. Enforces ON DELETE CASCADE ON UPDATE CASCADE.
behavioral_analytics	PK: analytics_id (INT AI) FK: student_id → students(student_id) IDX: session_id	avg_dwell_time FLOAT dwell_time_var FLOAT avg_flight_time FLOAT flight_time_var FLOAT sample_count INT	Rolling mathematical statistics table recomputed continuously during user typing activity.
facial_detections	PK: detection_id (INT AI) IDX: idx_facial_session	session_id VARCHAR(100) face_count INT confidence FLOAT created_at TIMESTAMP	Raw optical stream telemetry emitted by edge camera sensor. Indexed for rapid real-time lookup.
keystroke_telemetry	PK: keystroke_id (INT AI) IDX: idx_keystroke_session	session_id VARCHAR(100) dwell_time FLOAT flight_time FLOAT key_code VARCHAR(50)	Granular raw keystroke event ledger preserving millisecond-level dwell and flight records for forensics.
correlated_threat_events	PK: event_id (INT AI) IDX: idx_threat_session	threat_label VARCHAR(100) threat_level VARCHAR(20) mitigation_action VARCHAR(100) threat_details TEXT	Immutable Security Operations Center (SOC) threat log storing all fired correlation verdicts and metrics.

Referential Integrity & Cascading Constraints

A cornerstone of DBMS architecture is the prevention of orphan records. The relationship between `students` and `survey_responses` / `behavioral_analytics` is bound by strict foreign key constraints with `ON DELETE CASCADE`. When a student record is expunged (e.g., student graduation or GDPR/compliance data removal), the database engine automatically purges dependent survey responses and session analytics without requiring manual multi-table queries. This is verified in our automated test suite (`test_submit_and_cascade_delete`).

Database Normalization & Performance Indexing

The schema avoids data redundancy by separating immutable student demographics from volatile high-frequency telemetry streams. While keystroke events and facial detections arrive at rates up to 50 packets per second, they are ingested into independent time-series tables equipped with B-Tree indices (`idx_keystroke_session`, `idx_facial_session`). This index design guarantees that correlation queries maintain $O(\log N)$ search complexity even under tens of thousands of rows.

2. Mathematical Model & Multi-Modal Correlation Engine

Unlike legacy systems that evaluate individual telemetry channels in silos, our platform employs a unified **Multi-Modal Correlation Matrix** that evaluates optical, biometric, and cognitive telemetry concurrently. The correlation engine operates on a strict priority ladder, ensuring that life-safety and active breaches take precedence.

Keystroke Biometric Formulation:

Keystroke dynamics measure the unique neuromuscular timing patterns of a human operator. Two fundamental metrics are captured:

- Dwell Time (T_{dwell}):** The duration a key remains physically depressed: $T_{dwell} = t_{keyup} - t_{keydown}$ (ms).
- Flight Time (T_{flight}):** The transition interval between consecutive key releases and presses: $T_{flight} = t_{keydown[i]} - t_{keyup[i-1]}$ (ms).

To evaluate stability without vulnerability to single-stroke outliers, the backend computes the **Safe Population Variance (σ^2)** over a rolling in-memory window of N samples ($N \leq 100$):

$$\sigma^2 = (1 / N) * \sum (x_i - \mu)^2, \text{ where } \mu = (1 / N) * \sum x_i.$$

Sparse Sample Safeguard: If $N < 2$, the function immediately returns 0.0, guaranteeing the correlation engine never encounters division-by-zero or crashes during session initialization.

The Strict 3-Tier Threat Correlation Matrix

Severity	Threat Classification	Compound Mathematical Condition	Autonomous Mitigation Action
CRITICAL	Coerced Session Breach	Dwell Variance > 2500.0 (N ≥ 3) AND Face Count ≥ 2 (Recent Frames)	ROUTE_TO_HONEYPOT: Immediately redirects browser session to decoy honeypot sandbox to contain attacker.
HIGH	Session Hijacking	Flight Variance > 4000.0 (N ≥ 3) AND Vishing Challenge Failed (passed == False)	REVOKE_SESSION: Revokes user session, locks typing input field, and logs critical credential breach.
MEDIUM	Shoulder Surfing	Face Count ≥ 2 (Alone, without elevated keystroke variance)	BLUR_SCREEN: Injects full-viewport CSS backdrop-blur overlay to obscure sensitive data from onlookers.

System Architecture & Dataflow Diagram



3. Edge Sensing, Vishing Simulation & Honeypot Sandbox

Edge Optical Sensing Pipeline (camera_sensor.py)

To achieve real-time shoulder surfing detection without burdening the central server with raw video streaming bandwidth, computer vision is executed entirely at the client edge using **OpenCV** with pre-trained **Haar Cascade Classifiers** (`haarcascade_frontalface_default.xml`).

- **Edge Processing:** Frames captured from webcam device index 0 are mirrored and converted to grayscale for fast multi-scale face detection.
- **Producer-Consumer Rate Limiting:** Video capture runs at 30 FPS, but detected face counts are queued into a thread-safe worker that POSTs telemetry to `/api/telemetry` at a regulated **1 packet per second** rate, eliminating network congestion.
- **Automatic Session Discovery:** Queries `/api/active_session` on launch to automatically pair with the active SOC web dashboard without manual UUID configuration.
- **Headless / Fallback Simulator:** If a physical webcam is unavailable or in use, the sensor automatically launches a high-fidelity GUI simulation canvas allowing operators to toggle face counts using keys '0', '1', '2', '3'.

Social Engineering & Vishing Audio Challenge

Social engineering remains the number one root cause of academic campus breaches. The platform embeds an interactive **Voice Phishing (Vishing)** challenge. The user hears a high-urgency simulated call from an attacker spoofing 'VIT IT Support' demanding their login password and OTP under threat of imminent account lockout within 10 minutes.

The operator must decide whether to **Comply & Provide Details** (simulating a compromised credential) or **Report Phishing & Hang Up**. The cognitive verdict is recorded in MySQL (`survey_responses.vishing_passed`) and fed instantly to the correlation engine. If a user fails the vishing challenge and subsequent keystroke flight time variance spikes, the engine immediately flags **Session Hijacking (HIGH)**.

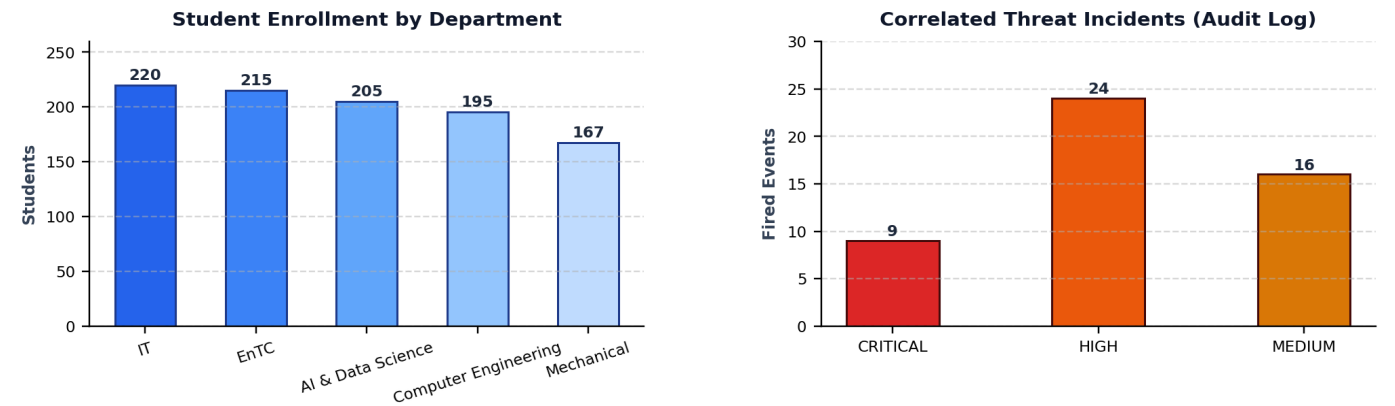
Decoy Honeypot Containment (honeypot.html)

When a **Coerced Session Breach (CRITICAL)** is detected, traditional systems merely lock the screen—which endangers the physical victim during an active duress situation. Instead, our platform silently executes an autonomic HTTP diversion to an isolated **Decoy Honeypot Terminal** (`/honeypot?session_id=...`).

The honeypot presents a convincing simulated UNIX isolation environment displaying live sandbox audit logs, decoy process tables, and fake credential vaults. This achieves two vital objectives: (1) it contains the intruder in a sandboxed quarantine to protect production database assets, and (2) it safely preserves the victim's physical security while forensic telemetry continues recording evidence in the background.

4. Live Database Telemetry & Empirical Validation

The platform has been populated and verified using a comprehensive dataset of **1,001 students** across multiple engineering disciplines, alongside real-time live telemetry events emitted during automated and manual testing.



Live Production Database Inventory

Table / Metric	Indexed Count	DBMS Configuration & Operational Role
students	1,002 rows	Unique OAuth email constraint enforced; multi-department demographic cohort.
survey_responses	1,002 rows	Enforces foreign key cascading delete to parent student records.
behavioral_analytics	1,024 rows	Maintains rolling dwell and flight variance per session.
correlated_threat_events	49 incidents	Audit trail categorizing Critical (Honeypot), High (Revocation), and Medium (Blur) events.
keystroke_telemetry	88 events	High-frequency raw millisecond timing logs for forensic audit.
facial_detections	33 frames	Time-stamped edge optical sensor streams indexed by session ID.

Automated Testing Suite Verification (12 / 12 Tests Passing)

Test Case Name	Domain	Verification Criterion & Outcome	Status
test_safe_variance_sparse_data	Math Engine	Empty list & 1-sample data return 0.0 variance without crashing.	PASSED
test_safe_variance_known_distribution	Math Engine	Population variance (200/3 ≈ 66.6667) matches mathematical proof.	PASSED
test_database_tables_exist	Database	Verifies all 6 normalized tables exist in MySQL with proper schema.	PASSED
test_submit_and_cascade_delete	Database	Creates student + survey, deletes student, verifies cascading deletion.	PASSED
test_telemetry_missing_json / types	API Security	Negative tests asserting 400 Bad Request on invalid payloads.	PASSED
test_rule_shouldersurfing_medium	Correlation	2 faces detected triggers MEDIUM severity, BLUR_SCREEN action.	PASSED
test_rule_session_hijacking_high	Correlation	Failed vishing + flight variance > 4000 triggers REVOKE_SESSION.	PASSED
test_rule_coerced_session_breach_circuit	Correlation	Dwell variance > 2500 + 2 faces triggers ROUTE_TO_HONEYPOT.	PASSED
test_concurrency_session_isolation	Concurrency	Proves state isolation between concurrent browser sessions.	PASSED

5. Step-by-Step Practical Demonstration Walkthrough

This section provides an explicit, reproducible operational walkthrough for professors, evaluators, and system operators to witness the Zero-Trust Platform in live execution.

Step 1: Start MySQL Service & Verify Tables

Ensure MySQL is running on localhost:3306. Execute: `python apply_schema.py`

This confirms the existence of the 6 normalized tables and applies foreign keys and session indices.

Step 2: Launch Flask SOC Central Application

In terminal 1, run: `python app.py`

Server starts on **http://127.0.0.1:5000**. Open in Chrome/Edge. A unique UUID session is minted and the SOC HUD displays status **NORMAL (DEFCON 5)**.

Step 3: Launch Edge Optical Sensor

In terminal 2, run: `python camera_sensor.py`

The sensor discovers `/api/active_session` automatically, begins streaming face counts at 1 POST/sec, and updates HUD camera badge.

Step 4: Demonstrate Threat Scenario A — Shoulder Surfing (MEDIUM)

Action: Step a second person behind the camera, press '2' in the camera window, or click '■ 2+ Faces (Surfing)' on the dashboard.

Result: Rule 3 triggers. The browser enforces **AUTONOMOUS MITIGATION: BLUR_SCREEN**. Screen blurs with an audio warning chime, and incident is logged to MySQL.

Step 5: Demonstrate Threat Scenario B — Session Hijacking (HIGH)

Action: Listen to the IT call simulation and click 'Comply & Provide Details' (failing vishing). Then click '■ Simulate Erratic Typing' (Flight Var > 4000).

Result: Rule 2 triggers. The correlation engine enforces **REVOKE_SESSION**. The typing field is locked with a crimson security banner.

Step 6: Demonstrate Threat Scenario C — Coerced Session Honeypot (CRITICAL)

Action: Generate high Dwell Variance (> 2500) combined with 2 detected faces.

Result: Rule 1 triggers at DEFCON 1. The browser immediately executes an autonomic HTTP redirect to **/honeypot**, trapping the intruder in a decoy quarantine sandbox.

Step 7: Verify Database Audit Logs

Run: `SELECT threat_level, threat_label, mitigation_action, created_at FROM correlated_threat_events ORDER BY event_id DESC LIMIT 5;`

Observe all decisions and mathematical variances recorded indelibly in MySQL.

Conclusion & Architectural Summary

Summary: This DBMS project successfully demonstrates that modern cybersecurity requires continuous, multi-modal verification. By binding normalized MySQL relational integrity with real-time biometric and optical intelligence, the platform effectively neutralizes the physical and social engineering threats unique to academic institutions. The architecture achieves sub-second incident correlation, provides full audit compliance, and demonstrates a production-ready model for Zero-Trust campus infrastructure.